

Proposed Solution Report: Watching, Player Feedback, and Search Visibility Enhancements

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I. PROBLEM STATEMENT

Although the platform already supports streaming, search, and watch history, the homepage and player still lack some small but important user experience features expected in modern streaming systems. In particular, users cannot quickly resume partially watched content, the player does not clearly communicate buffering state, and search interfaces do not show the number of matched titles.

II. PROPOSED SOLUTION

We propose three lightweight but high-value interface enhancements:

- Add a *Continue Watching* row on the homepage using watch history data.
- Add a player loading indicator while media is buffering or not yet ready.
- Add visible search result counts so users can immediately understand query output size.

III. SOLUTION DETAILS

A. *Continue Watching Row*

The homepage will fetch watch history alongside catalog data. Any movie with recorded progress but not marked completed will be shown in a dedicated *Continue Watching* row. This improves usability by helping users resume content directly from the homepage.

B. *Player Loading Spinner*

The video player will track buffering and readiness events from the HTML5 video element and HLS playback flow. While the stream is waiting or still initializing, a centered loading overlay will be shown. This gives immediate feedback and avoids the impression that playback is frozen.

C. *Search Result Count*

Both homepage-integrated search and dedicated search results will display the total number of matching titles. This improves search clarity and gives users quick feedback on query quality.

IV. EXPECTED OUTCOME

These changes are expected to improve usability without affecting the existing backend architecture or deployment model. The homepage becomes more personalized, the player becomes more informative during startup, and search becomes easier to interpret.

V. SCOPE

This solution is limited to frontend behavior and the reuse of existing watch history data. No database schema changes are required.